

Validation of a Computational Cocystal Prediction Tool: Comparison of Virtual and Experimental Cocystal Screening Results

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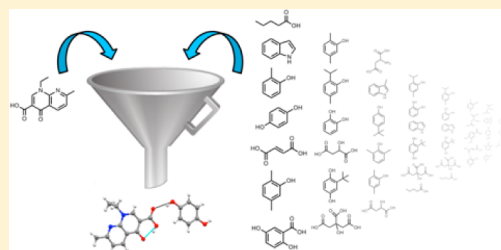
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S Supporting Information

ABSTRACT: A virtual cocystal screening method based on calculated gas phase molecular electrostatic potential surfaces (MEPS) of the individual components has been validated using experimental cocystal screens reported in the literature. The noncovalent interactions of a molecule with its environment are described by a discrete set of independent surface site interaction points (SSIPs), whose properties can be calculated from the *ab initio* MEPS. The stability of a crystal is estimated based on pairing SSIPs such that the sum of the pairwise interaction energies is optimized. This provides a means of calculating the relative stability of a cocystal compared with the pure components without knowing anything about the three-dimensional structures of the crystalline states. For a set of potential crystal coformers (CCF), the difference between interaction site pairing energies of different solid forms (ΔE) provides a method for ranking CCFs based on the calculated probability of cocystal formation. The method was applied to cocystal screens of 18 compounds that reported both hits and misses, and in most cases, the virtual cocystal screen reproduces experimental results well. In lists of CCFs ranked by ΔE , the experimentally observed hits were significantly enriched at the top, and this indicates that virtual screening is a promising tool for focusing experimental efforts on the most promising CCF candidates.



INTRODUCTION

Active pharmaceutical ingredients are materials that are most conveniently developed and delivered as solid dosage forms.¹ Pharmaceutical cocystals are known as molecular adducts of definite stoichiometry where one component is a neutral active pharmaceutical ingredient (API) and the other is a neutral counter molecule and both components are solids at room temperature.² Cocystals are composed of an API and a crystal coformer (CCF) held together by noncovalent interactions within the same crystal lattice.^{3,4} Pharmaceutical cocystals raise important intellectual and physical property issues in the context of drug development and delivery.⁵ By diversifying the number of crystal forms that exist for a particular API, cocystals can lead to improvements in physical and chemical stability as well as in mechanical properties. Cocystals of an API may show enhanced solubility, bioavailability, stability, and dissolution rates compared with the pure material.⁶

Cocystals have been prepared by cooling and evaporative techniques, as the methods of first choice,⁷ as well as sublimation, crystallization from the melt, slurries, grinding, and solvent-drop grinding.⁸ High-throughput (HT) screening experiments have been run over the past decade, successfully leading to novel cocystals of low solubility APIs.⁹ Because of the time and cost of experimental studies, complementary tools have been developed to increase the chances of cocystal formation in experimental screens. Predictive approaches to identifying molecular components likely to form cocystals are

based either on structural analysis using experimental data from the Cambridge Structural Database (CSD)^{2,10} or computational methods for calculating lattice energies.¹¹ The CSD has been used to identify frequently occurring supramolecular synthon pairs for use in cocystal design, and statistical analysis of cocystals found in the CSD has identified shape, size, and polarity descriptors for predicting the complementarity of cocystal components.¹⁰ *Ab initio* methods have been developed for calculating crystal lattice energies and predicting crystal structures, and these approaches have had some success in accounting for the formation of cocystals based on the difference between the calculated lattice energy of the cocystal and the sum of the calculated lattice energies of the pure components.¹¹

Here we use a computational method based on calculated molecular electrostatic potential surfaces (MEPS) to assess molecular complementarity and identify molecules that are likely to form cocystals. The approach is based on an electrostatic model that treats intermolecular interactions as point contacts between specific polar interaction sites on molecular surfaces.¹² The MEPS of a molecule is calculated in the gas phase, and this is used to identify a discrete set of surface site interaction points (SSIPs), which are described by

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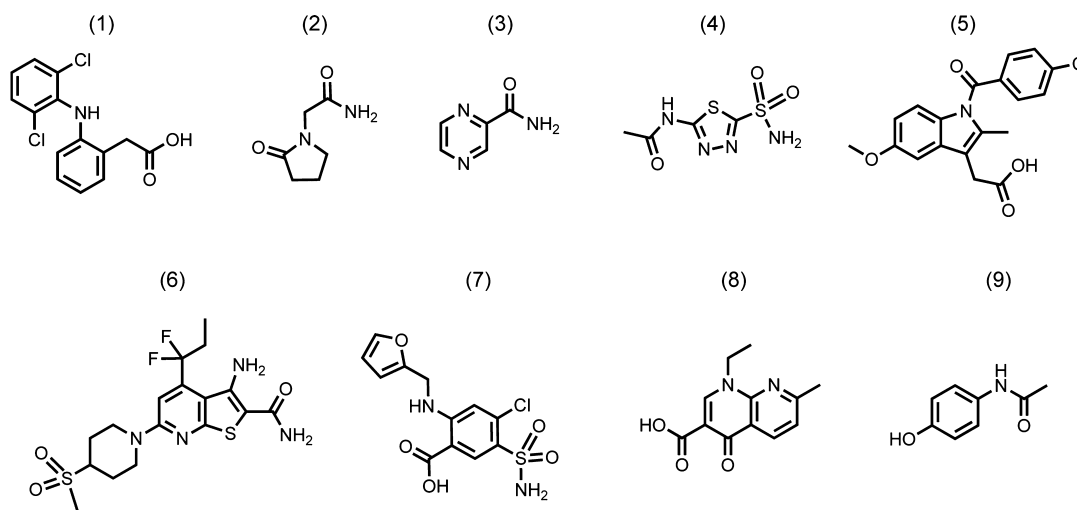


Figure 1. APIs used in experimental cocrystal screens: (1) diclofenac; (2) piracetam; (3) pyrazine carboxamide; (4) acetazolamide; (5) indomethacin; (6) drug candidate; (7) furosemide; (8) nalidixic acid; (9) paracetamol.

H-bond donor and H-bond acceptor parameters α and β .¹³ SSIPs identify conventional H-bond donor and acceptor sites as well as less polar sites that make weak electrostatic interactions, so they completely describe the surface properties of a molecule and can be used to calculate the total interaction of a molecule with its environment. The SSIP description has been shown to provide an accurate method for estimating solvation energies, the free energies of transfer of solutes between two different solvents, and solution phase association constants for formation of intermolecular complexes.^{13,14} We have also shown that summing the pairwise interactions between SSIPs in a solid can be used to estimate the probability of cocrystal formation between two compounds.¹⁵

In this approach, the probability of cocrystal formation is estimated by comparing the calculated energy of the cocrystal with the energy of the two pure phases. It is difficult to predict crystal structures accurately, which makes first principles calculations of lattice energies unsuitable for high throughput cocrystal screening. We therefore adopt an approach that does not require any knowledge of the details of crystal packing. Etter's rule suggests that the two most polar sites (strongest H-bond donor and acceptor) in a solid are likely to be paired (H-bonded) in the crystal.¹⁶ This principle can be extrapolated to construct a hierarchical list of all interactions in a solid. If the potential interaction sites (SSIPs) are ranked in order of polarity, H-bond donors and acceptors can be paired sequentially until no more pairwise interactions can be formed. This provides a straightforward method for determining which SSIPs will form intermolecular interactions in a crystal without any knowledge of the details of the three-dimensional structure. We can therefore estimate the stability of a solid using the interaction site pairing energy (E), which sums all contacts across the entire surface of all of the molecules present in the crystal (eq 1).

$$E = -\sum_{ij} \alpha_i \beta_j \quad (1)$$

where α_i are the positive SSIPs or H-bond donor sites and β_j are the negative SSIPs or H-bond acceptor sites. Pairings of the α_i and β_j SSIPs are chosen to minimize the value of E using the hierarchical scheme described above. Any excess positive or

negative sites that are left unpaired are considered not to contribute to the energy of the crystal.

A multicomponent solid is treated in the same way as a single component solid. Thus, the SSIPs of the cocrystal components are combined into a single list, and the best H-bond donors are sequentially paired with the best H-bond acceptors to obtain an interaction site pairing energy for the cocrystal. The difference between the interaction site pairing energies of the pure components and the corresponding energy for the cocrystal provides a measure of the thermodynamic driving force for cocrystal formation, ΔE (eq 2).

$$\Delta E = E_{cc} - nE_1 - mE_2 \quad (2)$$

where E_1 , E_2 , and E_{cc} are the interaction site pairing energies of the pure solids 1 and 2 and a cocrystal of stoichiometry $1_n 2_m$ respectively.

There are a number of assumptions and approximations implicit in this approach. For example, all interactions in a crystal are assumed to be attractive, so the lowest possible value for ΔE is zero. Three-dimensional structure and the arrangement of SSIPs in space are ignored, so individual SSIPs are considered to be independent and neighboring interaction sites do not affect one another. A single low energy conformer is used to calculate the SSIPs for each molecule. Conformation affects the number and the values of the SSIPs,¹³ but a low energy extended conformation describes the maximum interaction that a molecule can make with its environment.

We have previously applied this approach to virtual cocrystal screening of APIs against hundreds of potential CCFs taken from the GRAS list (generally regarded as safe additives by the Food and Drug Administration).^{15,17} Here, we test the validity of the approach using experimental data from the literature. Despite the large number of cocrystal screens that have been reported, most studies focus on the structures of successfully formed cocrystals, the hits, rather than including a list of all attempted cocrystallizations. For validation of prediction methods, the misses are just as important as the hits. Nine experimental cocrystal screens that report both the hits and the misses were used in this study.¹⁸

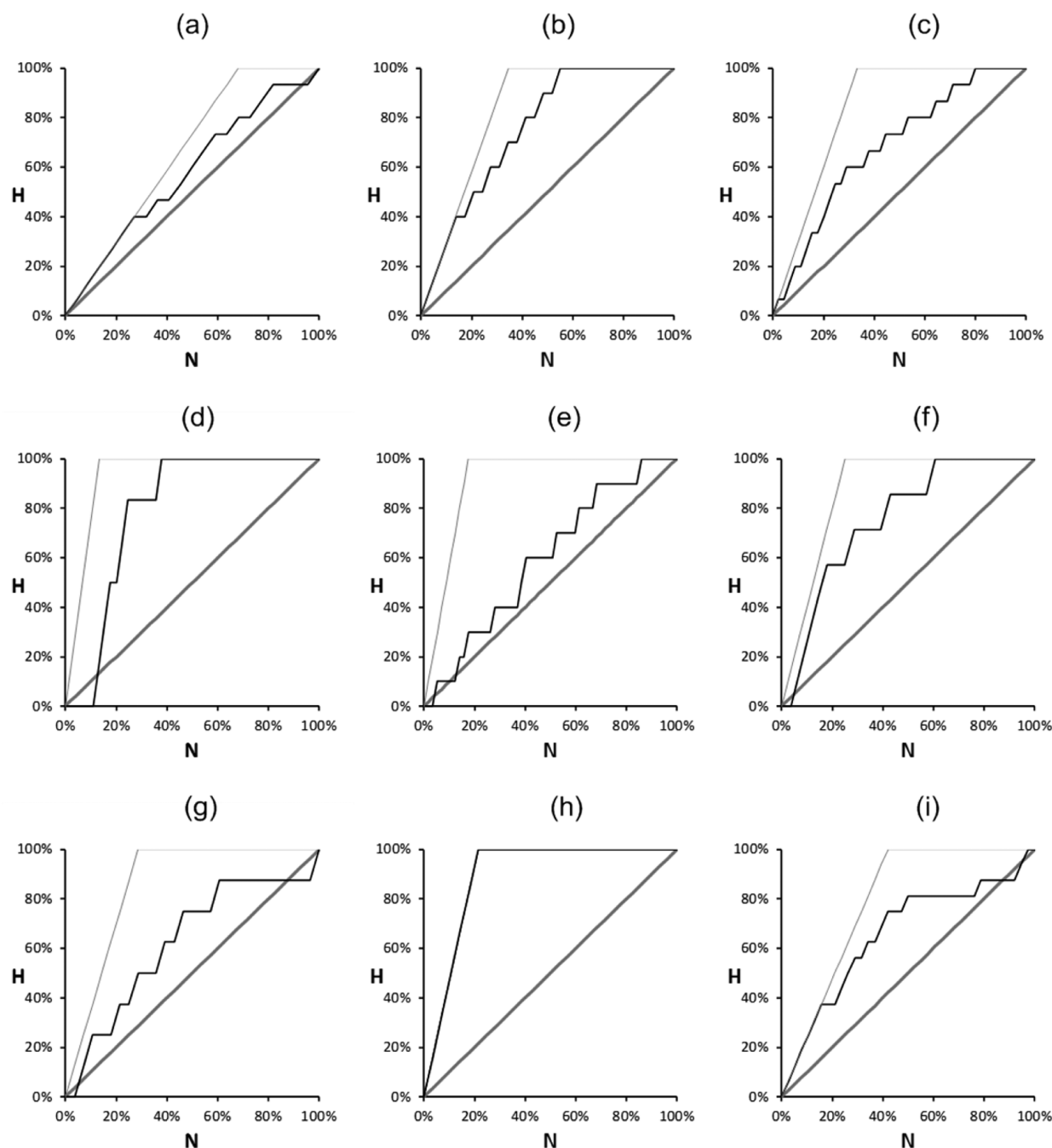


Figure 2. Recall plots for the prediction of cocrystal formation between (a) diclofenac (1) and 22 CCFs, (b) piracetam (2) and 29 CCFs, (c) pyrazine carboxamide (3) and 45 CCFs, (d) acetazolamide (4) and 36 CCFs, (e) indomethacin (5) and 57 CCFs, (f) drug candidate (6) and 28 CCFs, (g) furosemide (7) and 28 CCFs, (h) nalidixic acid (8) and 22 CCFs, and (i) paracetamol (9) and 37 CCFs. H is the fraction of hits found, and N is the fraction of compounds sampled. The thick gray lines represent the results that would be obtained if the hits were evenly distributed among the list of coformers, and the thin gray lines represent the results that would be obtained from perfect prediction. The black lines represent the results obtained based on the ΔE ranking.

RESULTS AND DISCUSSION

The nine APIs used in the experimental cocrystal screens are shown in Figure 1 (see Supporting Information for details of CCFs). The molecular structures were drawn in an extended conformation and energy-minimized using molecular mechanics.¹⁹ All structures were then energy-minimized using DFT, the MEPS was calculated, and this was converted into a set of SSIPs (see Supporting Information).¹³ For each cocrystal screen, the SSIPs were used to calculate the values of ΔE for all 1:1 API–CCF cocrystals using eqs 1 and 2.

For each cocrystal screen, the CCFs were ranked according to the value of ΔE , and this ranking was used to construct recall plots, which compare the predicted probability of cocrystal formation with the experimental results. In cases where a CCF was experimentally found to form a salt with an API, we included this CCF as a hit, because it is sometimes difficult to differentiate the extent of proton transfer due to the salt–cocrystal continuum.²⁰ The recall plots in Figure 2 show the fraction of hits found descending the ΔE ranked list, H , plotted as a function of the fraction of compounds sampled, N (black

line). The thick gray line represents the result that would be obtained if the hits were evenly distributed among the list of cofomers, where the probability of cocrystal formation is independent of the number of compounds screened. The thin gray line represents the result that would be obtained from a perfect prediction, where the hits were all at the top of the ranked list. In all cases, the calculation gives a significant enrichment in hits near the top of the ranked list. In the case of compound 8, nalidixic acid, the computational method correctly identifies all six CCFs at the top of the ranked list. The recall plot for each cocrystal screen in Figure 2 can be condensed into a single score (S , see Experimental Section), which quantifies the enrichment of hits at the top of the ranked list relative to an even distribution of hits among the list of cofomers: the thick gray lines in Figure 2 correspond to no enrichment ($S = 0$), and the thin gray lines correspond to 100% enrichment with all of the hits at the top of the ranked list ($S = 1.0$). The results in Table 1 show that the scores are

Table 1. Scores (S) and p -Values for Mann–Whitney U Tests (%) for Virtual Cocrystal Screens Compared with Experimental Results

compd	S	p	no. of CCFs screened	no. of hits found
1	0.49	3.9	22	15
2	0.78	0.0	29	10
3	0.55	0.1	45	15
4	0.68	0.4	36	6
5	0.23	7.5	56	10
6	0.69	1.1	28	7
7	0.35	10.6	28	8
8	1.00	0.0	22	6
9	0.54	0.3	38	16

significantly greater than zero for all of the virtual screens, which indicates that using the virtual screening approach is superior to randomly selecting compounds from the list of potential CCFs. Six of the nine screens have scores higher than 0.5, and compound 8 gave a perfect prediction ($S = 1.0$).

The Mann–Whitney U test was used to establish whether the enrichment of hits toward the top of a ranked list is statistically significant, and the corresponding p -values are given in Table 1. If $p < 5\%$, we can be 95% confident in the ability of the ΔE values to discriminate between hits and misses. The values in Table 1 show that this is the case for seven of the nine screens. The two screens with values of $p > 5\%$ correspond to the screens that have values of S lower than 0.4.

A related approach to cocrystal screening has been reported using the COSMO-RS software to calculate the difference in excess enthalpy (H_{ex}) between the two pure components and the cocrystal. This method assumes that interactions in the solid phase are equivalent with a supercooled liquid phase.²² Figure 3 shows the 11 APIs used in this study (see Supporting Information for details of CCFs).²³ For some of these systems, the number of CCFs screened was similar to the number of hits, which limits the value for validation of computational prediction methods. The computational methods described above were used to calculate ΔE for these systems, and the results are compared with the experimental results in Figure 4 and Table 2. Note that some of the compounds in Figure 3 are the same as those shown in Figure 1, but for the purposes of comparison with the COSMO results, the recall plots in Figure 4 and the scores in Table 2 only use the CCFs reported in ref 22. In the indomethacin screen, Raman spectroscopy indicated that tromethamine and *N*-methyl-*D*-glucamine form salts. These CCFs were included as hits in the analysis below, and the neutral forms of the compounds were used in the calculations.^{23h}

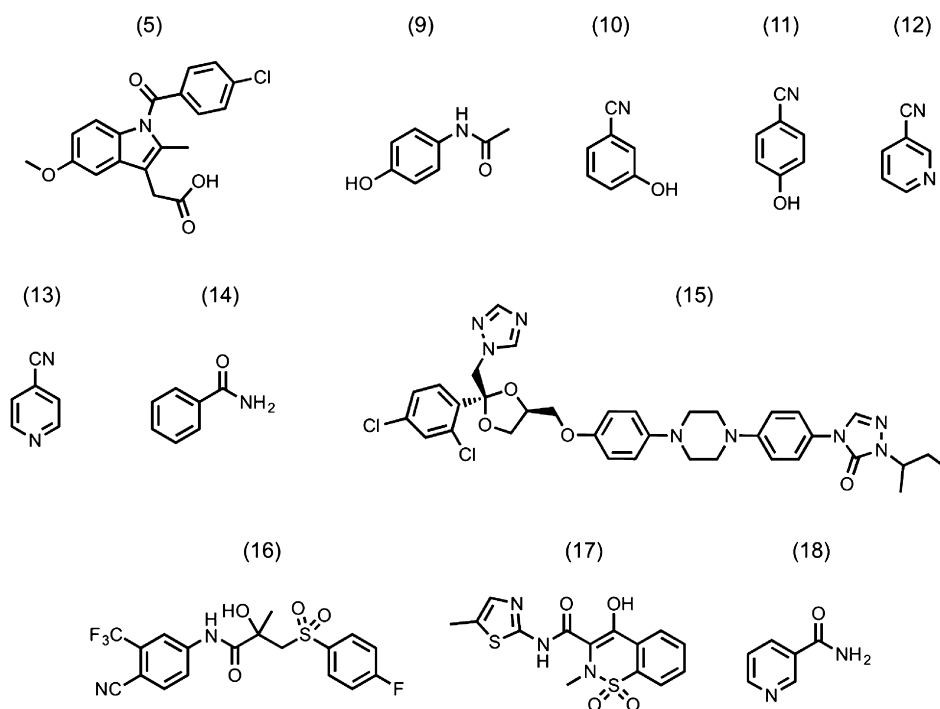


Figure 3. APIs used in experimental cocrystal screens that were previously studied using COSMO: (5) indomethacin; (9) paracetamol; (10) 3-cyanophenol; (11) 4-cyanophenol; (12) 3-cyanopyridine; (13) 4-cyanopyridine; (14) benzamide; (15) itraconazole; (16) bicalutamide; (17) meloxicam; (18) nicotinamide.

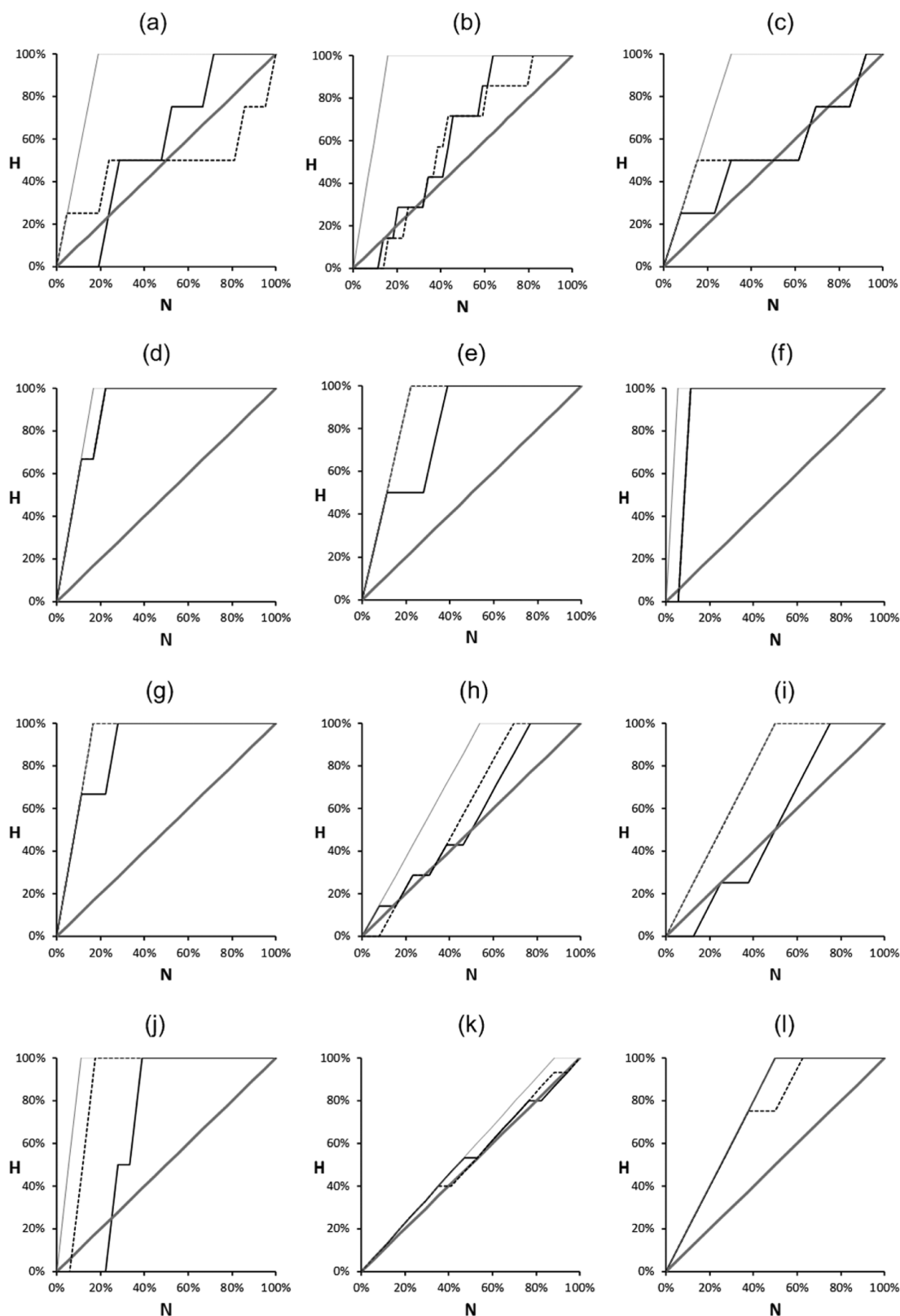


Figure 4. Recall plots for the prediction of cocrystal formation between (a) indomethacin (**5**) and 21 CCFs; (b) indomethacin (**5**) and 44 CCFs; (c) paracetamol (**9**) and 13 CCFs; (d) 3-cyanophenol (**10**) and 18 CCFs; (e) 4-cyanophenol (**11**) and 18 CCFs; (f) 3-cyanopyridine (**12**) and 18 CCFs; (g) 4-cyanopyridine (**13**) and 18 CCFs; (h) benzamide (**14**) and 13 CCFs; (i) itraconazole (**15**) and 8 CCFs; (j) bicalutamide (**16**) and 18 CCFs; (k) meloxicam (**17**) and 17 CCFs; (l) nicotinamide (**18**) and 7 CCFs. *H* is the fraction of hits found, and *N* is the fraction of compounds sampled. The thick gray lines represent the results that would be obtained if the hits were evenly distributed among the list of coformers, and the thin gray lines represent the results that would be obtained from perfect prediction. The solid black lines represent the results obtained based on the ΔE ranking. The dotted black lines represent the results obtained based on the COSMO ranking.

Table 2. Scores (*S*) and *p*-Values for Mann–Whitney U Tests (%) for Virtual Cocystal Screens Compared with the Experimental Results Reported in Reference 22

compd	<i>S</i> (ΔE)	<i>S</i> (COSMO)	<i>p</i> (ΔE)	<i>p</i> (COSMO)	no. of CCFs screened	no. of hits found
5 ^a	0.21	−0.03	37.7	55.3	21	4
5 ^a	0.26	0.20	13.8	21.1	44	7
9	0.11	0.22	41.3	30.2	13	4
10	0.96	0.96	0.3	0.3	18	3
11	0.79	1.00	0.9	0.0	18	4
12	0.88	0.88	11.1	11.1	18	1
13	0.91	1.00	0.5	0.1	18	3
14	0.29	0.43	22.3	11.7	13	7
15	0.13	1.00	10.0	1.4	8	4
16	0.44	0.87	23.5	2.9	18	2
17	0.33	0.33	30.9	26.5	17	15
18	1.00	0.83	2.9	5.7	7	4

^aTwo different experimental screens of indomethacin are considered.

Figure 4 compares the COSMO predictions (dotted black line) with the interaction site pairing energy predictions (solid black line), the expectations for an even distribution of hits in the ranked list (thick gray line) and perfect prediction (thin gray line). In most cases, both computational methods show a significant enrichment of hits at the top of the ranked list. The two virtual screening methods perform extremely well for compounds 10, 11, 12, 13, and 18 with scores above 0.75 for both methods (Figure 4d,e,f,g,l). COSMO performs significantly better than ΔE for compounds 15 and 16 (Figure 4i,j). For compounds 5, 9, 14, and 17, the two virtual screening methods give only a small enrichment of hits near the top of the ranked list. However, for compound 17, there are 15 hits and only 2 misses, which leaves a very small window for enrichment. Overall, there is little to choose between the two virtual screening methods with average scores on these data sets of 0.53 and 0.64 for the ΔE and COSMO methods, respectively.

The *p*-values for Mann–Whitney U tests are also given in Table 2, and the values are higher for these data sets than for those reported in Table 1. This means that our confidence in being able to discriminate hits and misses is significantly lower for the screens on the compounds in Figure 3. For the COSMO calculation, only five of the 12 screens give *p*-values less than 5%, and for the ΔE calculation, four of the 12 screens give *p*-values less than 5%. In line with the values of *S*, the *p*-value results are very similar for the two computational methods, but COSMO performs significantly better for compounds 15 and 16, and ΔE performs marginally better for compound 18.

CONCLUSIONS

An approach to virtual cocystal screening has been validated using literature data on experimental cocystal screens of 18 APIs. The approach assumes that the interactions of a molecule with its environment can be described by a relatively small number of surface site interaction points (SSIPs).¹³ The properties of the SSIPs for individual molecules were calculated from the gas phase *ab initio* MEPS using a footprinting algorithm described previously.^{13b} Summing pairwise interaction energies between SSIPs was used to estimate the total intermolecular interaction energy of a solid (*E*).¹⁵ Comparison

of the interaction site pairing energy of a cocystal with the corresponding energies of the two pure components (ΔE) provides a means of assessing the probability of cocystal formation. For each experimental screen, the CCFs were ranked according to the values of ΔE and the results were compared with the experimental data. In most cases, the calculations found that the experimentally observed hits had large values of ΔE , and these compounds were found near the top of the ranked list. Comparison of this computational approach with an alternative virtual cocystal screening method based on COSMO showed little difference between the two methods.²² Although there are some instances where the virtual screen failed to make accurate predictions, in general, the results show that calculation of ΔE provides an excellent method for selecting promising CCFs for experimental screening with an API.

EXPERIMENTAL SECTION

The molecular structures were drawn in an extended conformation in TorchLite Version 10.0.0, Revision 18308.¹⁹ The structures were energy-minimized using the XED3 force field in TorchLite, and geometry optimizations were then performed using DFT (B3LYP 6-31G*) in Gaussian 09. The DFT MEPS was converted into a set of SSIPs,^{13b} and these were used in eqs 1 and 2 to calculate the difference in interaction site pairing energy for cocystal formation, ΔE . For each cocystal screen, the CCFs were ranked based on the values of ΔE (see Supporting Information for details). These ranked lists were used to construct the recall plots shown in Figures 2 and 4.

It is also possible to calculate values of ΔE for cocystal stoichiometries other than 1:1. However, the ranked list obtained for 1:1 cocystals is practically identical to that obtained for different stoichiometries (see Supporting Information), so only 1:1 stoichiometries are reported in this paper.

The value of ΔE can also be converted into a probability of cocystal formation, as described previously (eq 3).¹⁵

$$P = \frac{e^{-(\Delta E + 11)/(RT)}}{1 + e^{-(\Delta E + 11)/(RT)}} \quad (3)$$

where 11 kJ mol^{−1} is an empirically determined constant that disfavors cocystal formation.

The recall plots were constructed by defining a series of points with coordinates (*N*(*i*), *H*(*i*)) for the *i*th CCF in the ranked list. Given a total of *N*_{CCF} CCFs, the values of *N*(*i*) are given by *i*/*N*_{CCF}. The result expected if the hits were evenly distributed in the ranked list was calculated using values of *H*(*i*) given by *R*(*i*) = *i*/*N*_{CCF} (thick gray lines in Figures 2 and 4). Given a total of *N*_H hits, the result expected for a perfect prediction was calculated using values of *H*(*i*) given by *P*(*i*) = *i*/*N*_H for *i* < *N*_H and *P*(*i*) = 100% for *i* > *N*_H (thin gray lines in Figures 2 and 4). The recall plot for the virtual screen was calculated using values of *H*(*i*) given by *V*(*i*) = *h*(*i*)/*N*_H, where *h*(*i*) is the cumulative number of hits found on descending the ranked list as far as the *i*th CCF (black and dotted lines in Figures 2 and 4). The overall score (*S*) for a virtual screen is defined using eq 4.

$$S = \frac{\sum_{i=1}^{i=N_{CCF}} V(i) - \sum_{i=1}^{i=N_{CCF}} R(i)}{\sum_{i=1}^{i=N_{CCF}} P(i) - \sum_{i=1}^{i=N_{CCF}} R(i)} \quad (4)$$

The statistical significance of the enrichment of hits toward the top of a ranked list was evaluated using the Mann–Whitney U test. One-sided *p*-values for this test were obtained using the ranksum function implemented in Matlab (version R2013a from www.mathworks.com).

ASSOCIATED CONTENT

Supporting Information

Comparison of the values of ΔE for different stoichiometries of paracetamol cocystals, CCFs used in each cocystal screen, and

the ranked list of ΔE values. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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